

LAB 4 REPORT

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Lab Title: User defined function

Theory: User Defined Functions in Python are functions created by the programmer to perform specific tasks. They help in organizing code, reducing repetition, and making programs easier to understand. A user-defined function is created using the `def` keyword and can be called whenever needed.

```
. # TASK 1
```

```
def greet():  
    print("welcome to uet lahore")  
greet()  
greet()
```

Output:

```
welcome to uet lahore  
welcome to uet lahore  
  
Process finished with exit code 0
```

```
# TASK 2
```

```
def displayName(name):  
    print("Hello",name)  
displayName("SOHAIB")
```

Output:

```
Hello SOHAIB  
  
Process finished with exit code 0
```

```
# TASK 3
```

```
def add2numbers(num1,num2):
```

```
    return num1+num2
result = add2numbers(25,52)
print("The Result is:",result)
```

Output:

```
The Result is: 77

Process finished with exit code 0
```

TASK 4

```
def square(num):
    return num*num
result = square(5)
print("The result is:",r)
```

Output:

```
The result is: 25

Process finished with exit code 0
```

TASK 5

```
def power(base,exponent):
    return base**exponent
print(power(5,3))
```

Output:

```
125

Process finished with exit code 0
```

TASK 6

```
def student(name,age):
    print("Name:",name)
    print("Age:",age)
student(name="SOHAIB",age=18)
```

Output:

```
Name: SOHAIB  
Age: 18  
  
Process finished with exit code 0
```

```
# TASK 7
```

```
def maximum(a,b,c):  
    return max(a,b,c)  
result=maximum(20,25,52)  
print("The Result is:",result)
```

Output:

```
The Result is: 52  
  
Process finished with exit code 0
```

```
# TASK 8
```

```
def total(*numbers):  
    sum_numbers = sum(numbers)  
    print("Total:", sum_numbers)  
total(11, 2, 34)  
total(1, 52, 4)  
total(3, 5, 7)
```

Output:

```
Total: 47  
Total: 57  
Total: 15  
  
Process finished with exit code 0
```

```
# TASK 9
```

```
def average(*numbers):  
    avg=sum(numbers)/len(numbers)
```

```
print("Average:",avg)
average(1,2,34,52,7)
Output:
```

```
Average: 32.8
```

```
Process finished with exit code 0
```

```
# TASK 10
def add(num1,num2):
    return num1+num2
def sub(num1,num2):
    return num1-num2
def mul(num1,num2):
    return num1*num2
def div(num1,num2):
    return num1/num2
print("1.Add")
print("2.Subtract")
print("3.Multiply")
print("4.Divide")
num1 = int(input("Enter a number: "))
num2 = int(input("Enter another number: "))
choice = int(input("Enter your choice: "))
if choice == 1:
    print((add(num1,num2)))
elif choice == 2:
    print((sub(num1,num2)))
elif choice == 3:
    print((mul(num1,num2)))
elif choice == 4:
    print((div(num1,num2)))
else:
    print("Invalid choice")
```

Output:

```
1.Add
2.Subtract
3.Multiply
4.Divide
Enter a number: 25
Enter another number: 52
Enter your choice: 1
77

Process finished with exit code 0
```

